RURAL FARMERS RESEARCH NETWORK

EXPLORING FARMER PRACTICES AND PERSPECTIVES ON USING TEPHROSIA FOR INTERNAL AND EXTERNAL LIVESTOCK PARASITE TREATMENTS

Findings from Focus Group Discussions



Prepared by:

Michael Chariga

January 2025

******* THIS PAGE IS INTENTIONALY LEFT BLANK**********

Table of contents

Table of contents....1
Executive summary....2
Introduction....3
Objectives....4
Methodology....5
Key findings....6
Variation in Treatment Practices....6
Ratios and dosages:....6
Soaking time:....6
Factors influencing variation:....7
Perceived Effectiveness of Treatments....7
Effectiveness:....7
Negative side effects:....9
Challenges:....10
Farmer Preferences and Innovations....10
Preferences:....10
Innovations:....10
Conclusion and recommendations from the groups....11
Conclusion....11
Recommendations....11
Limitation....11
Annex....12

Executive summary

This Focus Group discussions (FGD) report is based on the findings from 6 focus group discussions with over 58 farmers who volunteered for a follow-up qualitative research. The discussions were conducted to better understand the findings derived from the analysis of the *December 2024 farmer Follow-up Survey* dataset. The FGDs were conducted in the 5 subcounties where RFRN has members. The participants in the discussions were from 6 targeted farmer groups namely; Naora, Kurutiyange, Suna East, Suna West,Nyatike and Uriri.

The report informs the *December 2024 farmer Follow-up Survey* which resulted to a report : Progress and Insight: Migori RFRN 2024 Farmer-Led Research on Using Indigenous Plants To Control Livestock Parasite to highlight and provide valuable insight on use of indigenous plants, specifically *Tephrosia vogeli* and neem leaves in controlling internal and external livestock parasites .

The key findings from the focus group discussions shows that tephrosia both powder and fresh leaves, are effective in controlling both internal and external livestock parasites. Farmers have expressed confidence in using tephrosia as alternative treatment to veterinary drugs. Farmers reported improved milk production, animal coat quality , high appetite and general good health of their livestock after deworming. However, there are still inconsistencies on ratio used for treatment and frequency of treatment. The majority of farmers use and advocate for using tephrosia treatment as a preventive treatment.

On the negative side, farmers noticed that ratio 1:3 had significant side effects on small livestock like goats,calves and sheep; these included excessive diarrhea that led to near death experience and itching and dizziness.

Farmers also noticed that storage of powder was a major determinant of its effectiveness and thus explaining why some farmers found ratio 1:5 not effective while others reporting it as highly concentrated

Introduction

The analysis of *December 2024 farmer Follow-up Survey ,* conducted in December 2024 as a farmer follow-up survey on the 2024 livestock research , highlighted key insights necessitating a second phase to gather more indepth qualitative data on effectiveness of various indigenous plants used in controlling livestock parasites. The survey revealed a variety of practices among farmers in preparing and applying the treatments:

- Treatment preparation: Different farmers used different ratios in preparing both internal and external treatment while others did not use ratio at all.
- Variability in deworming dosage: There were different measurements used to gauge the deworming dosage .
- Treatment interval: There was inconsistency in the interval at which the treatments were applied.
- Treatment soaking: The soaking time for both internal and external treatments varied widely.
- Preparation methods: Ratios and preparation methods also varied.

In addition, there were reports of *Tephrosia Vogeli* having negative side effects especially on calves and young livestock; these included irritation and prolonged diarrhea after deworming.

The survey findings revealed further that although farmers who attended the 2024 research planning did have an extension training sessions for those who did not attend, these trainings, as reported by majority of farmers, were not specific to the planned research. Although this could partially explain the observed discrepancies in treatment preparation and application, it does not fully account for all the variations.

Given these findings, it became necessary to investigate other factors that contributed to variations in research implementation, and to better understand and interpret the survey findings and address the challenges that led to variation in research by the Rural farmers Research Network .

Objectives

The FGDs were designed to build upon insights from the *December 2024 farmer follow-up survey.* Specifically, they aimed to:

1. **Understand variation in treatment practices:** The FGDs sought to explore why farmers used different ratios, dosages, and soaking times in preparing and applying treatments. This also included Investigating the underlying factors influencing these variations in preparation methods and application intervals.

2. **Examine the Perceived Effectiveness of Treatments:** Gather farmers' qualitative assessments of the effectiveness of indigenous plants specifically and neem leaves which were reported in the *December 2024 farmer Follow-up Survey* as the main plants used in controlling internal and external parasites. The FGDs also aimed to discuss the perceived benefits and challenges, including negative side effects on calves and young livestock.

3. **Gather Insights on Farmer Preferences and Innovations:** Understand farmers' preferences for treatment methods and their reasons for choosing specific ratios or intervals and todDocument any innovative practices farmers may have developed and their effectiveness.

Methodology

Between January 13th and January 17th, 2025, six focus group discussions (FGDs) were conducted with 58 farmers from five subcounties in Migori where the Rural Farmers Research Network (RFRN) has member farmers.

The FGDs were held in each subcounty, with one FGD conducted per subcounty, except for one subcounty that had one group split into two thus two FGDs

Farmers were invited to participate voluntarily through announcements made by subcounty coordinators and lead farmers. Each discussion group was composed of a mix of men, women, and youths from different clusters.

Each FGD was facilitated by the AE hub team using a semi-structured guide tailored to explore key themes identified from the December 2024 survey. Discussions were conducted in both local and swahili language language to encourage open participation and were recorded with participants' consent for later analysis.

Key findings

Variation in Treatment Practices

Ratios and dosages:

Farmers reported starting with ratio 1:10 of tephrosia powder because they thought it could be poisonous to the livestock but changed gradually to a less concentrated one. However, because of the information shared in the whatsapp groups from those who had started earlier, some farmers reported that they started with ratio 1:7 or 1:5 based on the information that they got about ratios that were effective.

> *"When I realized ratio 1:10 was not effective, I tried ratio 1:5 of tephrosia powder and the following day I saw the result. The cow had started showing signs of a rough coat but after two days the coat went back to normal. What happened was diarrhea and then it was okay afterwards" - Nelson Okaka (RFRN farmer from Nyatike).*

500ml was used for deworming mature cows and was reported effective across the sub counties. However, it was noted that the same measurement had a negative effect on calves and young animals.

> *" What I realised is that if you give calves the same dosage as a mature cow, 500ml, it experiences dizziness"- RFRN farmer, Suna West.*

Although it is clear that there should be different ratios used for livestock of different ages, size and type, there is still a challenge for some farmers who prefer allowing animals to drink directly from the basin.

While many farmers reported using ratio for preparing treatment ratio and measuring quantity for dosage, sentiments from some farmers especially the elderly ones from both Nyatike and Uriri revealed the opposite; these farmers reported that they were not using specific measures instead using handfuls to approximate the treatment and adding water and instead of using bottles for deworming, they put the water in the basin for animal to drink and adding local salt lick *(Balaa)* to encourage the animal to drink.

Other farmers reported using measurements to prepare the treatment, however, these measurements were not appropriate, for instance, one farmer reported using one cup of tephrosia with two litres of water.

Ratio 1:10 was noticed to be ineffective for both internal and external parasites. Ratio 1:5 worked best according to the majority in controlling both internal and external parasites for both cow and young animals. However, for internal parasites in small livestock, the majority reported that ratio 1:7 was effective with between no to mild negative consequences. Ratio 1:3 was observed by some farmers and perceived by the majority as risky for use in livestocks, both for internal and external treatment- with near-death experiences when used as internal treatment.

Soaking time:

Farmers from Nyatike argued that the soaking time for fresh tephrosia is different from that of the powder. They argued that fresh leaves need to be soaked for a short time like about three hours and not overnight and that it should only be used for external treatment. Their hypothesis was that the longer the fresh leaves are soaked, the more the chemical content is realised and thus very dangerous for internal

treatment. However, on the contrary, farmers from other subcounties reported soaking for 12 hours and others up to 24 hours but did not observe any negative side effects except for ratio 1:3 when used on small livestock.
Some farmers, majority of whom were elderly women, reported no soaking at all or soaking for some hours or minutes when using both tephrosia powder and fresh leaves.

Factors influencing variation:

Elderly women were reported not using any treatment measurements; instead, they used handfuls or just poured the treatment on the basin for the cow to drink.
Different farmers noted that the storage of tephrosia powder greatly influenced the effectiveness. For instance,they argued that a well stored (stored in airtight container away from sunlight) tephrosia powder was stronger and more effective compared to fresh leaves. Thus farmers who reported different ratios to be working well might have been greatly influenced by the treatment storage e.g. some farmers reported that ratio 1:5 of powder was very strong and recommended ratio 1:7 as the best while others recommended that ratio 1:7 while effective, was not as ratio 1;5 and thus were recommending 1:5 instead. All these could be valid based on the treatment storage as discussed by farmers.
There were farmers who used tephrosia as a preventive, i.e. not waiting for disease outbreak, for these groups of farmers, there was consistency in treatment application period e.g. some spraying for one month others two months. On the other hand, there were a group who said they were using the treatment when they observe symptoms, for this group, their interval of treatment varied e.g. 1 month, two or even three months

> *"I have not dewormed my cow again because I have not seen signs of disease of internal parasites"*
> - Rose Akumu Ndera, RFRN farmer from Nyatike

Perceived Effectiveness of Treatments

Effectiveness:

Farmers across the 5 sub counties reported observing mucus in feces, appetite, coat colour and worms in the feces as a checklist for treatment effectiveness. Farmer who used tephrosia powder for deworming reported increase in milk production:

> *"I have five cows, I had been milking them and getting milk except for one . When it first gave birth it produced very little milk, the second birth the same. One of my friends told me the cow might have a problem. I wanted to sell it because I was keeping it but not getting any produce in return but she insisted that the problem could be resolved . She advised me to use 'Minya'- a local plant. But i was skeptical that it could kill my cow so i used tephrosia powder but added a small quantity of 'Minya' then natural salt lick (Balaa) then put in a basin about two litres. The cow drank it all. I tried the same thing in the evening . I also tried on the calf and the following day the calf was active. When I went to milk it the following day I got increased milk and the quantity continued to increase every day. I have now planted more tephrosia because I realized it is good."-RFRN farmer, Nyatike*

Fresh leaves were effective for treatment external parasites ; the observed results included making the skin smooth, removing spots on the skin and swelling in the stomach

" I had a dog that was frequently scratching itself so i decided to use tephrosia i pounded the fresh leaves and added water then thoroughly rubbed the dog with the pounded leaves and tried again after three days the dog stopped scratching itself and the skin became smooth"-RFRN farmer, Uriri

Across the 5 sub counties, farmers were confident that tephrosia, both fresh and powder, was effective in controlling ticks with many farmers reporting that ticks were dropping the same day or could take up to 4 days to disappear, however, they reported that the reinfestation period was quick with farmers saying that their livestock were reinfested nearly a week after spraying. From the majority of the farmers, tephrosia works best as a preventive not a curative treatment. Some farmers reported that they were consistently spraying frequently and as such were able to control the ticks.

Notably, even with many farmers reporting negative side effects of tephrosia fresh leaves on goats and calves, some farmers found tephrosia powder effective in deworming goats. For instance three farmers from Nyatike reported improved goat coat and diarrhoea stopping after using powder:

> *" The goat was having a rough coat and its diarrhea contained mucus so I took tephrosia powder in the evening in the ratio 1:10 and soaked overnight then sieved and used a 300ml bottle and dewormed the goat. I saw a slight change . After one week I tried the same powder but different ratio 1:7 and used 300ml dosage. After 3 days the diarrhea stopped and the feces were hard and after one week the coat went back to normal."*- Edwin Ouma, Nyatike

Farmers believed that tephrosia does not only control the parasites but instead had other hidden benefits that they had not yet confirmed.

> *" If you prepare and use tephrosia very well, it doesn't just work on one thing, it has multiple purposes on the cow"*- Denis Odongo, RFRN farmer from Uriri.

In support of Denis odongo's comment, other farmers, three from Nyatike and and four from Kuria west also observed that tephrosia treatment when used for deworming provided conditions where other disease causing factors could not thrive in the livestock body thereby boosting the livestock immunity.

A section of farmers used tephrosia fresh leaves when there was a lumpy skin disease outbreak in their neighbourhood and were able to both treat and control the disease from attacking their livestock.

> *"My cow started developing lumpy skin disease; there were lumps on the skin so i went and picked fresh tephrosia leaves,pounded then rubbed on the lumps until it turned reddish after two days repeated again ; did this thrice and the lumps dried and my cow is now healthy"*- David, Uriri farmer.

David's comment on tephrosia treating lumpy skin was also shared by another woman who also when she realised there was a lumpy skin disease outbreak in the neighbourhood used tephrosia as a preventive and her cow was not infected.

There is generally good perception about tephrosia treatment and positive adoption among both RFRN members and neighbour farmers. Farmers perceive it as a cheaper alternative to veterinary drugs that they use to buy from agrovet.

> *"The treatment is good, many people speak highly of it and it is what we use on chickens and cattle that are safe. Now we have alternative treatment so that even when there is disease outbreak we don't run to the agrovet, we have now planted more tephrosia. ..our chickens are no longer attacked by mites when brooding".*- Isdora Adhiambo, Suna west

Isdora's comment was also echoed by farmers from across the 5 subcounties with farmers from both Naora and Suna east reporting having recruited more farmers who joined after learning and seeing the positive impact of using tephrosia treatment.

Other than the benefits of tephrosia on internal and external livestock parasites, many farmers from across all the 5 sub counties , also told stories of how tephrosia treatment has become integral part of their chicken keeping. Sentiments that were constantly repeated in almost all the focus group discussions was how tephrosia has helped the farmers control chicken parasites and prevent chicken loss whenever there are disease outbreaks in the neighbourhood.

> *"Mites infested my chicken sleeping area, we agreed with my wife to pound fresh tephrosia leaves then mixed with water and left to soak for 3 hours. After three days all the mights were gone"- RFRN farmer, Nyatike*

> *" When I heard that there was a disease outbreak in the neighbourhood, I started putting tephrosia in the chicken drinking water. I realized that the outbreaks in my neighbourhood did not reach my chicken. My chicken grew in good health then i knew tephrosia was effective"-RFRN farmer, Naora*

However, as they share the benefits they have reaped from using tephrosia in chicken keeping, one thing stood out: tephrosia was more effective for those farmers who used it as a preventive measure, with other farmers who had not been consistent regretting loss of bird to diseases.

Negative side effects:

Farmers from Nyatike subcounty strongly believed that tephrosia fresh leaves could be harmful if used for internal parasites. One of the statements that was said by Samson Ouma and supported by four other farmers present was:

> *"When I was starting the experiment I took fresh leaves and pounded and soaked overnight and gave my calf . The calves diarrhea to a point of death. So I realised there is a difference between powder and fresh leaves. For me as a farmer, using fresh leaves to deworm is something that I don't trust because of how my calve diarrhea"*

The sentiment of tephrosia causing diarrhoea and even excessive scratching on animals was reported across the subcounties with another farmer in Kurutiyange also giving testimony on the near death experience when he used tephrosia to treat a goat.

Generally, the negative side effect was observed on young livestock like goats, sheep, and calves. And this was reported when farmers used ratio 1:3 and especially with tephrosia fresh leaves.

> *"I sprayed ratio 1:3 on one of my cows but it turned out that the coat started having patches and i sensed that this was not good for my cow, it can kill also when i used ratio 1:3 for deworming the cow experienced excessive diarrhea until i was forced to look for alternative ways to stop the diarrhea. For sheep, if you are not keen you can lose your sheep"-One farmer from Uriri sub county*

Challenges:

Some farmers expressed interest in knowing what exact diseases tephrosia treat; this question came up in both Naora and Nyatike sub counties. Although farmers acknowledged that they are currently treating signs and symptoms, they pointed to the need to document exact diseases that tephrosia cures.

> *"We should know specific diseases that tephrosia can treat so that we don't have a generalization that it treats every disease."- Joseph, farmer Naora group in Kuria west*

There is general concern to find a value addition for tephrosia powder so that its effectiveness is not short.

> *" If there are ways of preparing the powder so that it can take up to three months without expiring, that would be good instead of preparing powder that can only be used once"*- Uriri farmer.

It is difficult to know whether a given ratio is effective or not when used on animals that are not showing any signs of infestation.

> *"It is easy to know whether a given ratio is working or not if you use an animal that is already showing symptoms . However, if you use treatment on a healthy animal, it may be hard to notice changes."*-RFRN farmer, Suna east.

Ratio 1:7 and 1:5 are very effective in controlling external parasites but the problem is that the ticks reinfect the livestock after a short time thus needing frequent application.

There was concern that the same sprayer farmers use for applying tephrosia treatment were also used for spraying other chemicals in the farm.

> *"The challenge we have is that the spray we use is also used for different chemicals. If we could get a spray dedicated to use tephrosia only then maybe the observations could be different"*- Martin Olwal, farmer Uriri

Farmer Preferences and Innovations

Preferences:

Farmers across the five sub counties preferred using tephrosia as a preventive treatment, although a section of farmers used the treatment only when they observed signs of infestation.

> "If you want to be a wise farmer, don't wait until your cows fall sick. Tephrosia doesn't treat, it prevents. In tephrosia we practise prevention is better than cure"- Samson Ouma, Nyatike

Some farmers preferred using a piece of soap or washing powder to act as a sticker and reported that it was effective both in controlling the ticks and keeping the flies away.

Some farmers prefer pounding the fresh tephrosia leaves and using it as a paste instead of soaking

Innovations:

Elderly women from Uriri and Nyatike sub counties reported that instead of using bottles to forcefully deworm their cow, they added natural salt lick(*Balaa*) to lure the cow and drink in the basin.

Using tephrosia drops on the animal eye to control eye worm

> *"Tephrosia deworming does not easily remove eye worms so in order to remove them add some drops in the animal eye."*- RFRN Farmer, Suna west

Conclusion and recommendations from the groups

Conclusion

Across the six focus group discussions, it is clear that tephrosia both powder and fresh are effective in controlling livestock parasites both internal and external. Also farmers, both RFRN members and neighbour farmers are positively receiving and adopting the treatment of use in their livestock keeping. The effectiveness of tephrosia greatly relies on the preparation, dosage and storage(for powder). The variation in treatment ratio was highly associated with treatment storage with many farmers prefer applying tephrosia fresh leaves paste directly on the infected area. Farmers' creativity in adapting tephrosia treatment to their needs, such as adding salt lick to encourage drinking or adding soap as a sticker indicates a strong willingness to integrate tephrosia into their livestock keeping practices.

Recommendations

Explore and promote methods for extending the shelf life of tephrosia powder. These could include coming up with a value addition process for tephrosia powder or training farmers on drying techniques, airtight packaging and storage.

Recognize and support farmer innovations, like adding natural salt lick (Balaa) for internal treatment or adding soap as a sticker for external sparing, by incorporating them in formal recommendations.

Co-create with farmers a standardized measure for ratio and dosage. For example, instead of using ratios like 1:10, come up with a measure that farmers who use handfuls can rely on e.g. three handfuls of tephrosia powder in a 2kg tin full of water for treatment preparation.

Invest in further research to identify the specific livestock diseases tephrosia cures and establish scientific-backed practices for its use. Additionally, document the impact of tephrosia on livestock, both negative and positive, to guide future intervention.

Strengthen farmers whatsapp and the watano groups to facilitate pear learning and sharing of experiences. These groups could also be used to conduct trainings that enhance farmers knowledge on tephrosia preparation, storage and application.

Limitation

Participation in these FGDs was completely on a voluntary basis. As a result, there is a possibility of selection bias, where the findings may reflect the experiences and opinions of more active or interested farmers while potentially excluding insights from less engaged participants. This voluntary nature could also influence the diversity of the data collected, as certain demographic groups or farming practices might be underrepresented

Annex

Table 1: Details of group participants in the FGDs

Sub county	***Men***	***Women***	***Youths***	***Total***
Nyatike	*6*	*2*	*2*	**10**
Uriri	*5*	*3*	*4*	**12**
Suna East	*2*	*8*	*1*	**11**
Suna West	*2*	*3*	*3*	**8**
Naora	*5*	*3*	*3*	**13**
Kurutiyange	*3*	*0*	*3*	**6**
Total	**23**	**19**	**16**	**58**